



Warehouse Management System

PennWest California

User Manual

CSC 4900 – Senior Project 1

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Project Overview and Application

This project is a warehouse management system which can create efficient, organized, and responsive warehouse operations by using applications like bootstrap, Apache, docker, Visual studio code, PHP, MySQL and JavaScript. Our project is a web application that uses MySQL as our database, to then break it into rows and columns for easy data retrieval and storage. PHP is the main programming language that allows for same file HTML, PHP, and JavaScript coding. This means PHP can generate dynamic page content, its efficient at creating, reading, updating and deleting data and can interact with our database to store, retrieve and manage data within our database based on user input. This helps optimize processes like inventory tracking, order fulfillment, space utilization and shipment management.

By maintaining real-time records and automating repetitive tasks, reducing errors will enhance overall warehouse performance. This will ensure that companies can manage their supply chains more effectively and stay competitive in the market. The project will utilize Bootstrap as the front-end framework to build a responsive interface, AJAX and JavaScript will be used to improve user interaction, Apache will be used for our bridge between our web application and the internet as our web server and Docker will provide us a container to run all our applications in one place to be able to interact with each other. For our simplified management, a streamlined warehouse operation was implemented, introducing role-based access to ensure data integrity and secure management of products.

Our applicant will be able to handle automating inventory updates, stock management, and reporting. Along with being able to handle its products with the inclusion of QR and Barcode scanning for easy product management and auditing. How users will interact with our system is

through various pages on our dashboard, consisting of storage and products, orders, inventory, vendors, customers, returns, users and roles.

Motivation:

The design and overall structure of our web application is to make people's lives easier in the warehouse management space along with there is a big market for the system we developed. When discussing our design, we wanted to help warehouse managers efficiently track and manage their inventory; this includes ongoing shipments, deliveries, and supplies within the warehouse. While existing systems are available, they are not as affordable or rely on manual or outdated systems to track stock, leading to inefficiencies, errors, and increased operational costs. The team aimed to make a warehouse management system to simplify inventory tracking and management.

The main aspiration when designing our application was to focus on user-friendliness, compatibility, and practical use by warehouse managers and workers as most warehouse managers lack these functionalities that we provide. This includes a responsive design for warehouse managers and workers to minimize the learning curve to use our system. Our system includes QR and Barcode scanning for easy product management and for viewing within the warehouse. Overall, the team wanted to construct a cool and effective system for others to use.

Comparison to existing similar products:

There are quite a few warehouse management systems out there in the world, each of them having strengths and weaknesses. Fishbowl Warehouse is a WMS that provides functionality like SMB-friendly inventory management, automated reorder points, barcode scanning and tracking and multi-location inventory management. In our system we integrate a lot of the functionality that is other system has along with having them both being cost-effective for SMBs and easy to use.

Also, Zoho Inventory has features like budget-friendly, multi-channel order tracking and barcode scanning for quick fulfillment.

For local businesses the best WMS options are these two managers mentioned above, as they are cost-effective and easy to use but lack deep automation and AI. Knowing this, the team wanted the system to focus heavily on having a user-friendly interface, since our target audience may not have an IT team, barcode scanning and inventory tracking to reduce manual errors, order and shipment tracking for smooth fulfillment and cloud-based access so the user can manage inventory remotely. We also developed features for real-time inventory updates for users to always know stock levels, low-stock alerts to prevent shortages or stockouts, multi-location tracking so that the business can grow with our system since they would have access to multiple storage areas and customizable role base access control so that the user can have data integrity within the system. Our WMS is like other budget friendly systems, but different because of all the other functionality added to it to make it an all-in-one system for small businesses.

Community or social implications of the project

The positive impacts of our management system were one of the main driving forces for choosing this project. Some positive impacts are it would open new jobs in system management, IT support and logistics as our system would need to be maintained. There will be economic growth as our system will make for faster and more reliable supply chains which will benefit local businesses. Along with an enhanced consumer experience with the use of our efficient system, consumers will receive products quickly and correctly.

The implementation of our system will lead to a reduced environmental impact by optimizing inventory and shipping reduces our waste and carbon footprint. For all positives comes negatives and some negative impacts can be workplace stress. Since the WMS uses real-time tracking to monitor worker productivity this can lead to a high-pressure environment. An economic disruption can occur within small businesses that are not using our WMS and may struggle to compete with large, automated warehouses.

System block diagram

Project implementation details

Difference from design document:

The difference between this document and our design document is the construction of the document. Our design document is to give the development team a strong foundation to reference when developing this project. This will help aid and guide us through the functionalities and goals of this warehouse management system as well as keeping everyone on track without deviating too much from the source. That will include detailed descriptions of all

the software and functions needed for the project to work effectively and efficiently. Along with the audience being different since the design document is mainly for developers, engineers, project managers and designers. While this document is displaying all the functionalities implemented within the final project from the design document. This includes plain language, screenshots, UI walkthroughs and a step-by-step guide. This document is made for the user, customers, or operators to follow understanding the structure and functionality of our web application to then being updated based on user feedback and product updates.

Another way that these two documents differ is that the team has had to alter some of the designs shown in our design documents. An example of this is the fact that we included nothing about vendor and customer containers but are included because of further thinking of how we wanted to construct the project. This can also be shown on various pages with added functionality that was not included in the design document since when implementing our ideas, we came up with new innovative ideas to make our system work even better by having new functionality we didn't think of when designing our project.

The Gantt chart shown in figure gave the team a timeline at the beginning of the project. The team gave time slots to make sure there was enough time to implement all functionality required. There were many parts within the project that took longer than expected or required more thinking to move forward with cause for some implementations to be added before others. As well as there were new ideas like stated previously that were implemented in this phase that were not discussed in previous documents. This was offset by the fact that we started this project during the winter, so we got quite a bit done then. During this phase of the project, we have been

focusing on adding functionality each week that helps the user enjoy our system.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1		Winter Break	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14
2	Learn PHP, HTML, MySQL															
3	Create Environment/Repository															
4	Acquire Web Template															
5	Construct Database															
6	Create CRUD pages with core classes															
7	Create RBAC Functionality															
8	Implement Inventory & Products															
9	Create Customer Orders Functionality															
10	Implement Mathematical Calculations for various insights															
11	Testing															
12	Final Presentation															
13																

Challenges during implementation:

There were many challenges when constructing this project with some being bigger than others, but some small challenges were things like getting our environment set up for each developer in the team and for each developer to understand a new language. Along with the structure of our data tables and how data is going to be displayed. Even though these are small they were essential to help with the foundation of everything else in the project. Other bigger challenges were the implication of RBAC which is our role-based access control system. The complication with RBAC stemmed from the fact that we had to go back and add this functionality to each page and then test each page and the role to ensure that our system was working properly.

As well as getting vendor orders functionality created and the logic for utilizing the containers to associate with the vendor orders. When in development we realized that we would have to prioritize something else to process our orders which is why we added customer and vendor containers. These containers were important to separate vendors' and customers' orders so

that managers can distinguish which orders we are dealing with. Since this was not a part of our design, we had to pivot to which took a lot of time to develop and implement. So overall the main challenges were getting vendor orders functionality created. The logic for utilizing the containers to associate with the vendor orders, getting customer orders functionality created the logic for utilizing the containers to associate with the customer orders, reworking functions of customer orders for improved readability, the implementation of multiple modals for different issues and actions on the same page.

Use of software engineering principles

The team used the project construction standards discussed in previous documents during the Fall semester to develop our WMS. The project was broken down into the requirements, specification, design, and implementation.

The requirement phase was completed early in the project, the document contained information that was an overview and refined the team's project and its idea's. This was the beginning of our project and was the foundation of future documents. Some of the ideas within the document were changed or added throughout the project's progression, but the bases of the ideas remain throughout all our phases.

The next phase was the specification document which was completed mid-semester during the fall. This was a curial phase since it was necessary to define each process and how it would function with set software. The construction of this document gave us insight into how exactly each component would function and interact with each other. There are different flowcharts and block diagrams that were created to help visualize the process and future of the project.

We wrapped up the fall semester with and design document which provided a full breakdown of each module and went over each process needed for the project. This helped find and handle future logical errors and improved the overall structures of the program.

The implementation phase was set up by the previous phases and will be completed over the duration of the spring semester. It was important to refer to these documents so when the team began implementation, we had a guide to assist us through each function we wanted in the project. A Gantt chart was made to manage implementation time with each function added along with a changelog that held all our commits, which were tested during development and then pushed to our environment to then be added to the software.

User's Manual

Our Warehouse Management System was designed and developed with employees and employers that work in a warehouse in mind. This system has dynamic features for all its users, including automating inventory updates, stock management, and data security. Our application is a tool for managing and storing inventory within a warehouse allowing for businesses using it to grow.

Login

To login to the WMS application, the user must be added by their employer who hired them. The employer will need to direct themselves to the user's add page which is under Administration > Users > Add User shown in figure 2.1.

2.1 Warehouse Management System Interface

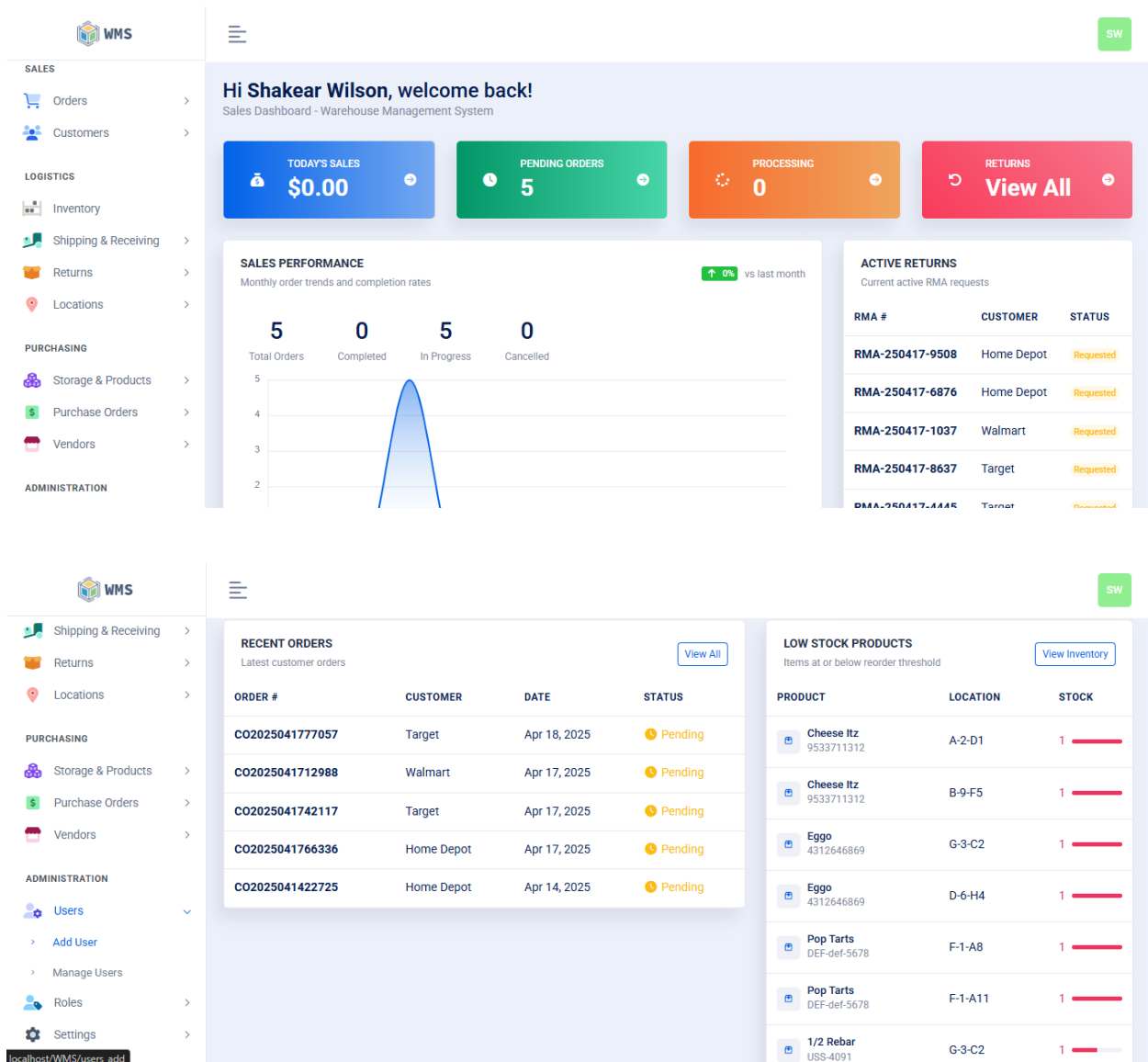


Figure 2.1 Warehouse Management System dashboard

2.2 User Login Information

The image displays a web application interface for adding a new user. It features a sidebar with various icons and a main form area. The form contains the following fields and controls:

- First Name ***: Text input with value "Demo".
- Last Name ***: Text input with value "Demo".
- Email ***: Text input with value "demo123@outlook.com".
- Username ***: Text input with value "demo123".
- Password ***: Password input with masked characters "*****".
- Confirm Password ***: Password input with masked characters "*****".
- Title**: Text input with value "demo".
- Country Code ***: Dropdown menu with value "United States (+1)".
- Phone Number ***: Text input with value "(412) 555-5511".
- Role ***: Text input with value "Warehouse Associate".
- Active Account**: Toggle switch, currently turned on.
- Buttons**: "Cancel" and "Register User".

The interface includes a sidebar with icons for navigation and a top right corner with a green "SW" button.

Figure 2.2 User Login Information administration view

Figure 2.2 shows the users add page on the employer's point of view since they are adding a new user to their system adding security. The information needed from the user is a first and last name, valid email address, username, password and a valid phone number. The administration will decide your title and role in the company and will activate your account.

2.3 User Login

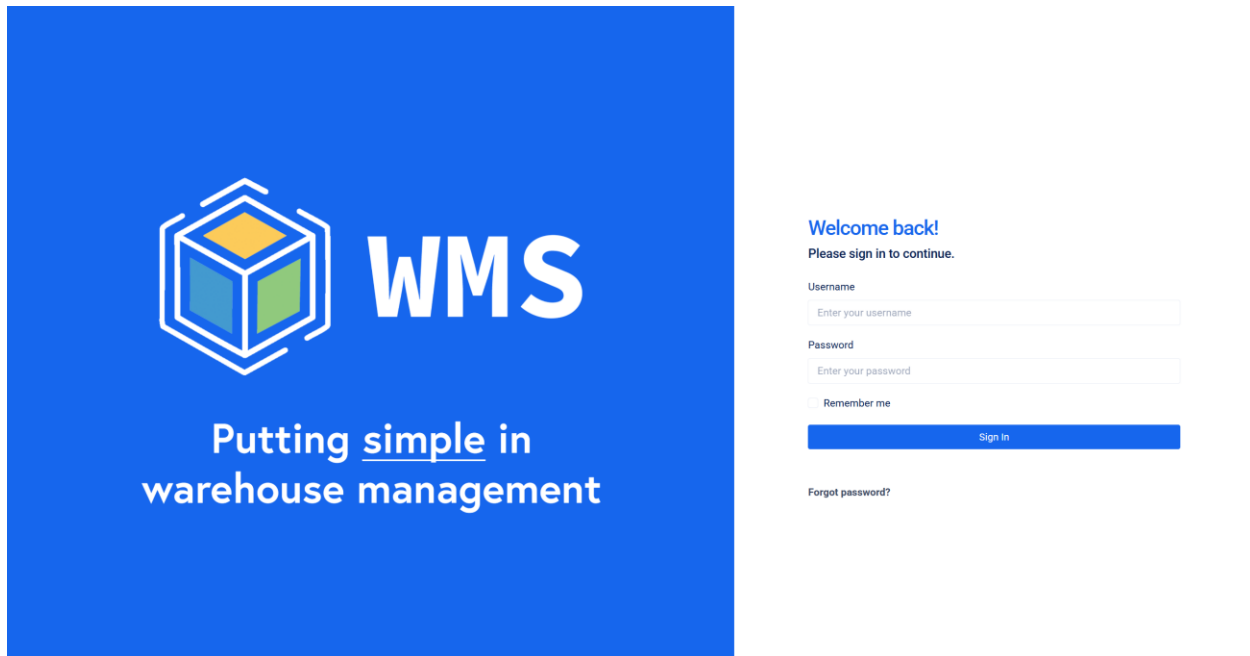


Figure 2.3 User Login user view

Figure 2.3 The user will then receive an email to activate their account, once the user clicks activate account on that email it will redirect the user to the login page. This display that your account is now active through email verification and now the user can login in with their username and password at: Login - <http://localhost/WMS/login>

Welcome

After the user logs in they will be directed to our main dashboard which displays statistics about the company and on the left-hand side shows the pages that are available. There are various pages the user can see and interact with, the user's access to some pages might be limited since they are based on a role. A role will grant you access to specific pages with features

needed for that set role, making other pages inaccessible for that user that don't involve set role. As stated previously, these roles are assigned by administration.

Transition to different type of roles and go in-depth on the feature that are on set page for that role and why they are important for that user to have.

Warehouse Assoicate

A warehouse associate plays a key role in a warehouse when dealing with incoming shipments, order fulfillment and warehouse organization. The associate will be involved in tasks like receiving and unloading shipments, inspecting and organizing inventory, picking and packing orders and potentially operating warehouse equipment.

3.1 Warehouse Associate

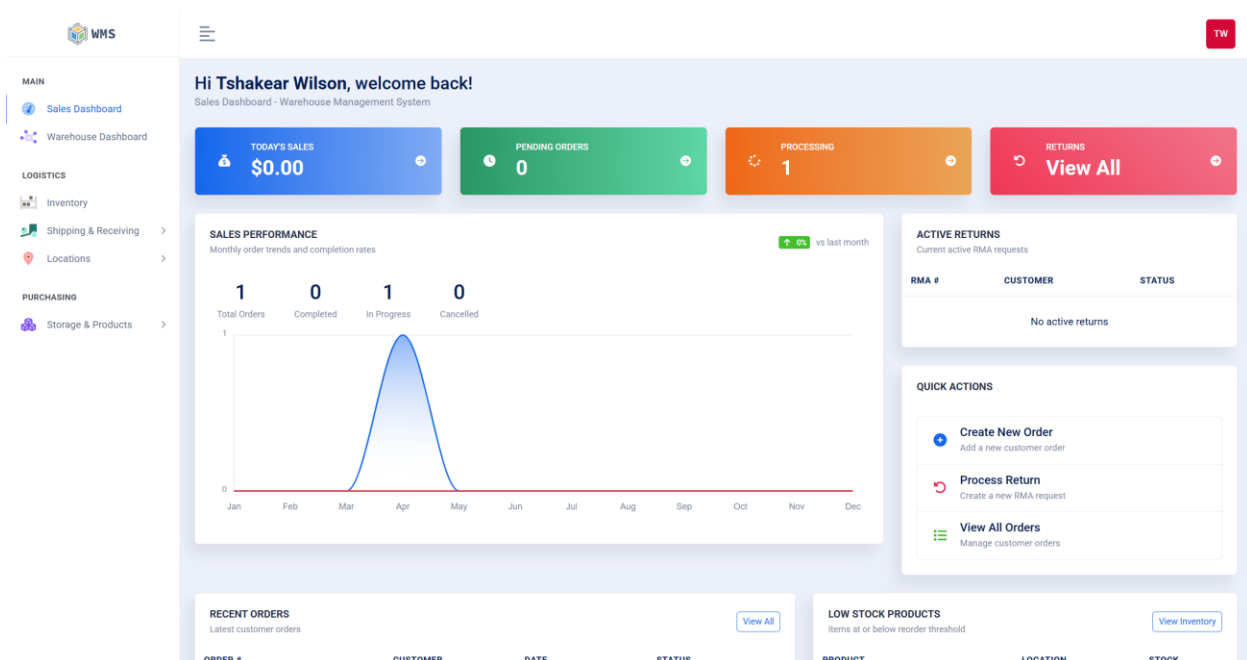


Figure 3.1 Warehouse Associate dashboard view

Figure 3.1 A warehouse associates permissions consist of a dashboard, inventory, shipping & receiving, locations and storage & products. These are the only pages a warehouse associate will have access to as these are the only pages needed for them to do their job efficiently.

3.2 Warehouse Associate Inventory

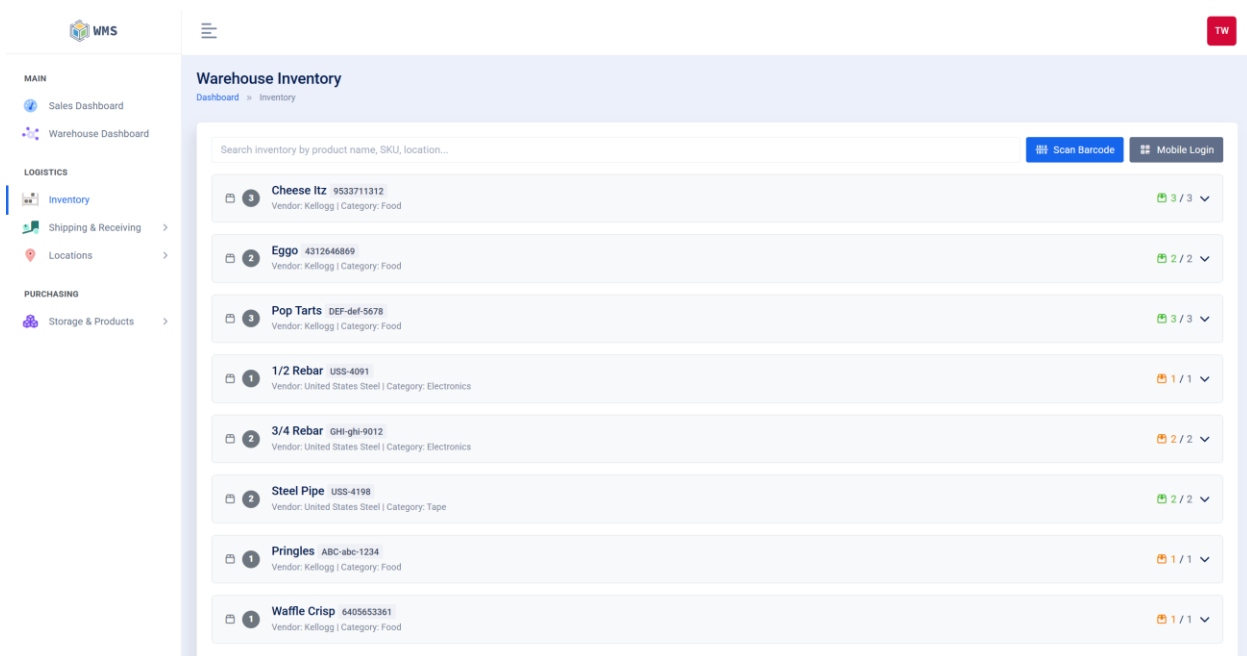


Figure 3.2 Warehouse Associate inventory view

Figure 3.2 This page has many different features and data that the user can interact with. There is a search field for the users to search inventory by product name, sku or locations. Other features on this page include barcode scanning, mobile login and printing out product labels which will be discussed later. The data shown on this page includes the products that are in the

warehouse inventory, the quantity of that product and the different containers that the product is being stored in.

3.3 Warehouse Associate Shipping & Receiving



Figure 3.3 Warehouse Associate Shipping & Receiving view

Figure 3.3 The warehouse associate will have two options under Shipping & Receiving, incoming and outgoing shipments. Incoming shipments are the products that are being delivered to the warehouse from our vendors. The process begins when an order is placed, and the vendor sends an advance shipping notice detailing what has been sent. Once the shipment has arrived, the warehouse staff checks those items and conducts an audit. Outgoing shipments deal with getting products out of the warehouse and sending them to a customer. The associate will pick and pack the products according to the order. The order will then be handed off to a delivery service for transportation.

3.4(a) Warehouse Associate Incoming Shipments

DATE	ORDER NUMBER	STATUS	VENDOR	ACTIONS
Apr 17, 2025	# VO2025041769042	Ordered	Kellogg	Resume Audit
Apr 17, 2025	# VO2025041764368	Ordered	United States Steel	Start Audit
Apr 16, 2025	# VO2025041664714	Completed	Kellogg	Audit Completed
Apr 15, 2025	# VO2025041527149	Completed	United States Steel	Audit Completed
Apr 14, 2025	# VO2025041415062	Completed	Kellogg	Audit Completed

Showing 1 to 5 of 5 entries

Previous 1 Next

Figure 3.4(a) Warehouse Associate incoming shipment's view

Figure 3.4(a) This page's features consist of order sort that is based on the status of the order, or the user can use the search field to search for a specific order. The user will have the option to resume an audit or to start an audit on an incoming order.

3.4(b) Warehouse Associate Incoming Shipments

Figure 3.4(b) Warehouse Associate incoming shipment's view

Figure 3.4(b) This page is where the associate will conduct their audit on a specific order. The information on this page consists of the product, product container, SKU and the quantity of the product that's coming in. Some features are a search field to search for products or the container along with barcode scanning functionality. The data that the user needs to input is to assign a customer container for set vendor product and to assign locations for all products in the order. Customer container equivalence that is shown in figure 3.4(c) shows how the user will select or search for a customer container, if the user uses auto-select equivalent then the system will try to find a container that fits the description. The same thing applies for universal location assignment and setting a location for a single product; the user can select or search for a location that's in the warehouse and assign that location to all products in that drop down by using apply to all items. After all items are processed then the audit can be completed shown in figure 3.4(d).

Customer Container Equivalence

This selection will apply to all items of this container [Auto-select equivalent](#)

Select a Container

Select a Container

(Crate) 3/4 Rebar (Retail)

(Bundle) 3/4 Rebar (Retail)

In (Unit) 3/4 Rebar (Retail)

(Box) Pop Tarts - Retail

(Crate) 1/2 Rebar (Retail)

2

1

Select Location

Edit

3

1

Select Location

Edit

✓	10		3/4 Rebar	Crate	SKU: USS-4091	▼
✓	5		3/4 Rebar	Crate	SKU: GHI-ghi-9012	▼
✓	30		3/4 Rebar	Bundle	SKU: GHI-ghi-9012	▼
✓	22		Steel Duct	Tray	SKU: USS-4200	▼
✓	16		Steel Pipe	Bundle	SKU: USS-4198	▼

Warehouse Associate Conclusion

The warehouse association also has access to locations including positions, levels, bays, racks and colors and storage & products pages that include vendor and customer catalog as well as vendor and customer containers. An associate work will not involve these pages, but they will still need access to them to view their data. They will only be able to view the page and will not be allowed to edit or add or delete these pages, the functionality and structure of these pages will be displayed in other roles.

Inventory Manager

An inventory manager in a warehouse plays a central role in ensuring that the flow of goods in and out of the facility is smooth, accurate, and efficient. Their job is all about control, coordination, and visibility over inventory from the moment products arrive to the moment they leave.

4.1 Inventory Manager

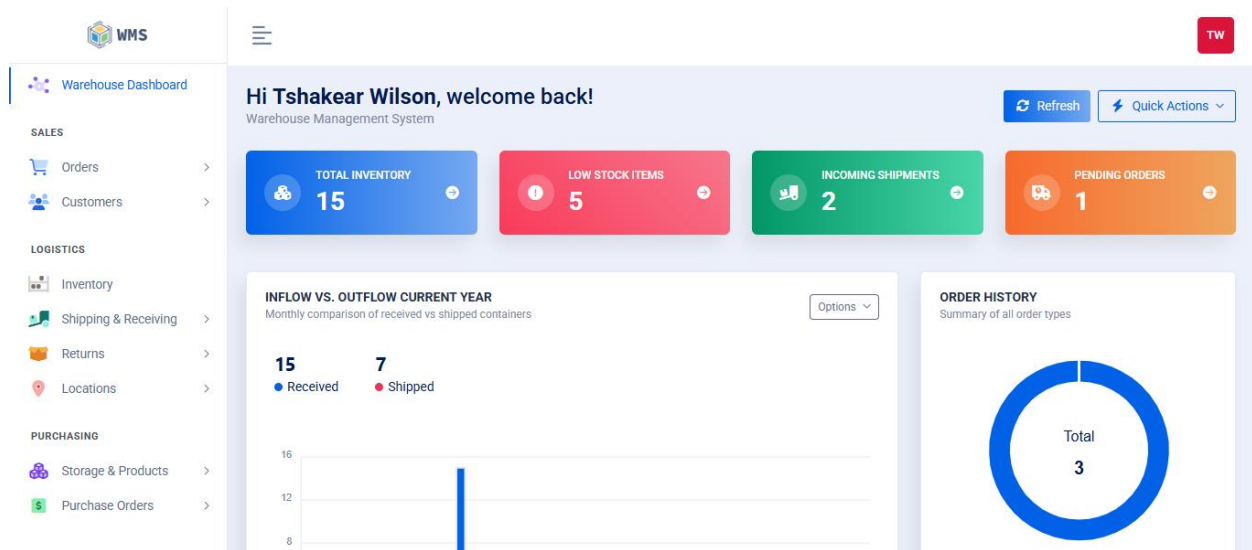


Figure 4.1 Inventory managers dashboard

In figure 4.1 the pages shown on the dashboard are all the pages needed for an inventory manager to do their job at a high level. Although these permissions look like a warehouse associate, there are many differences like the returns, purchase orders, orders and the features that are available in locations.

4.2(a) Inventory manager locations

WMS

Location Levels

Dashboard » Locations » Levels

+ Add Level Search...

LEVEL	RACK	BAY	COLOR	ACTIONS
A	F	1	Red	
C	G	3	Orange	
D	A	2	Light Blue	
E	Z	4	Pink	
F	B	9	Royal Blue	
H	D	6	Purple	

Showing 1 to 6 of 6 entries

LOGISTICS

- Inventory
- Shipping & Receiving
- Returns
- Locations
- Positions
- Levels
- Bays
- Racks
- Colors

PURCHASING

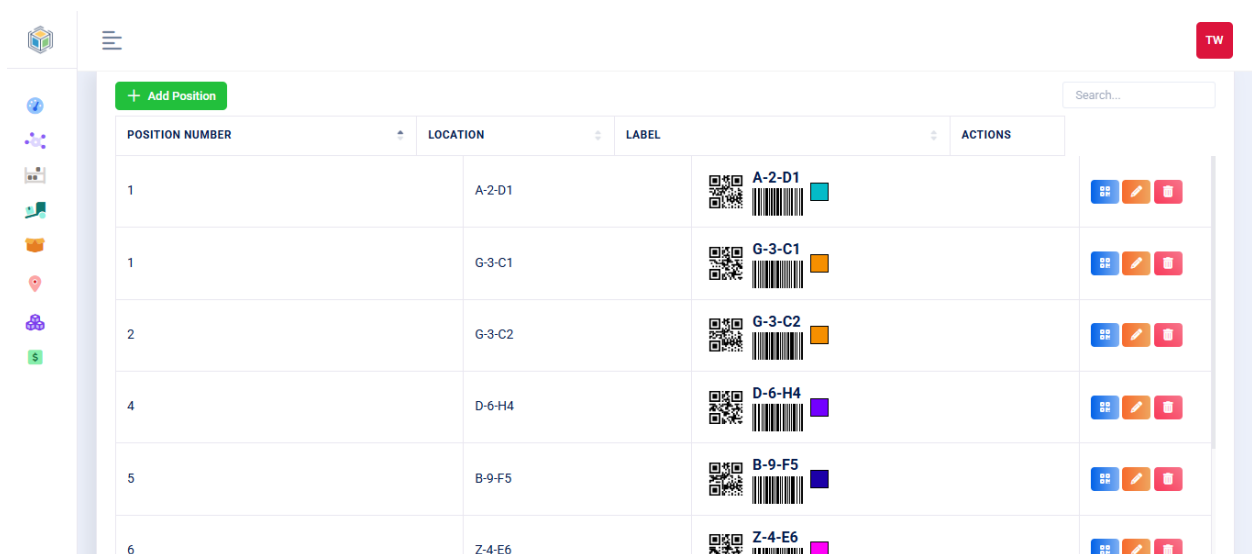
- Storage & Products
- Purchase Orders

TW

Figure 4.2(a) Inventory manager locations

Figure 4.2(a) shows how locations are going to be constructed throughout the warehouse. A location in the warehouse needs a rack, bay, level and color. When the inventory manager is done making the level then they can finish the locations by adding a position to it. These levels can be added, edited and deleted upon the user's request. There is also search functionality that is built into the page.

4.2(b) Inventory manager locations



POSITION NUMBER	LOCATION	LABEL	ACTIONS
1	A-2-D1	 A-2-D1 	  
1	G-3-C1	 G-3-C1 	  
2	G-3-C2	 G-3-C2 	  
4	D-6-H4	 D-6-H4 	  
5	B-9-F5	 B-9-F5 	  
6	Z-4-E6	 Z-4-E6 	  

Figure 4.2(b) Inventory manager locations

Figure 4.2(b) is the final step to creating a location and this is achieved by adding a location. This page has search functionality on it along with being able to print a location label, this label will hold information about the location. The user is allowed to add, edit and delete any location that they deem fit.

4.3 Inventory manager purchase orders

New Purchase Order
Dashboard » Purchase Orders » New Order

VENDOR INFORMATION
Kellogg

ORDER PRODUCTS

PRODUCT	CONTAINER	COST	QTY	TOTAL
Cheese Itz	(Level 1) Box	70.00	10	\$700.00
Eggo	(Level 1) Crate	225.00	15	\$3,375.00

ORDER SUMMARY
Subtotal: \$7,697.50
Tax: \$0.00
Total: \$7,697.50

ORDER DETAILS
Order Date: 2025-04-17
Order Status: Pending

Buttons: Submit Order, Cancel

Figure 4.3 Inventory manager new purchase order

Figure 4.3 shows that an inventory manager can create a purchase order which is important so that the inventory manager can reorder items that are low stock but will have to wait to get it approved from high ups. They can also view purchase orders to then conduct a return on them shown in figure 4.3(b).

Figure 4.3(b) Inventory Manager Returns

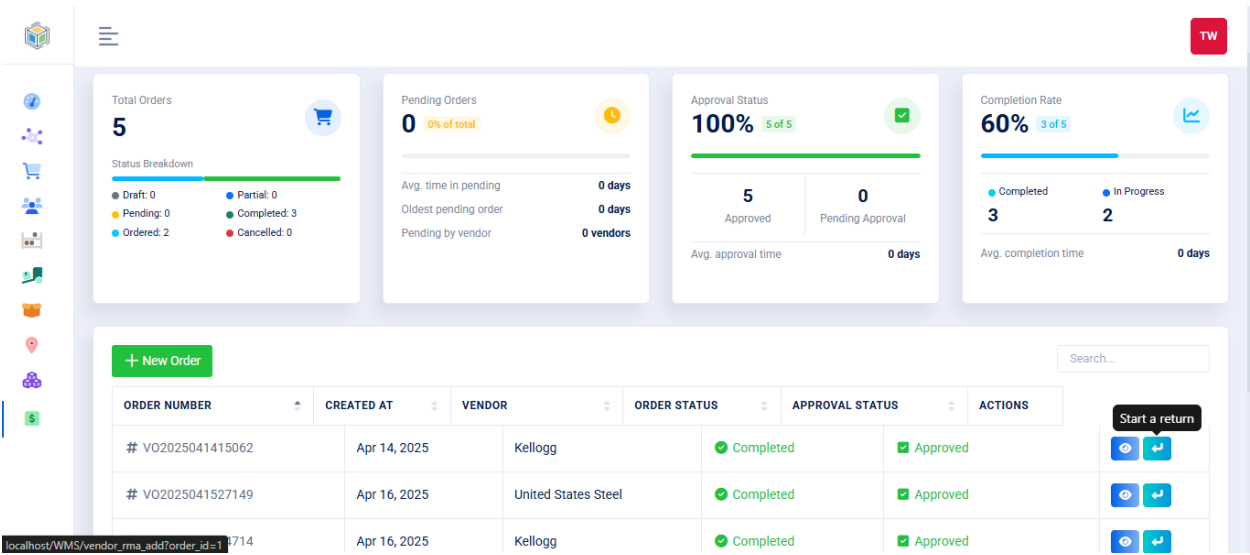


Figure 4.3(b) Inventory manager managing purchase orders

Inventory Manager Vendors Return

The screenshot shows the 'New Vendor Return' form. It includes a breadcrumb trail: Dashboard > Vendor RMAs > New Return. A 'Back to Orders' button is in the top right. The form is divided into two main sections: 'ORDER INFORMATION' and 'RETURN DETAILS'.

ORDER INFORMATION:

Vendor	Order Date	Order Status
Kellogg	Apr 14, 2025	Completed

RETURN DETAILS:

RMA Information:

- RMA Number ***:
- RMA Status ***:
- Requested Action ***:

Return Reason *:

Shipping Information:

- Shipping Carrier**:
- Tracking Number**:

Additional Notes:

Figure 4.4 Inventory manager vendors return page

This figure displays the information that is needed to start a vendor return along with some general information about the order. The information needed includes RMA number,

requested action, return reason and the selected products that will be returned, the user can also include the tracking number and shipping carrier. The requested actions can be a replacement product, a refund or store credit. The products also have a drop down for their conditions which can be used, damaged or defective, the reason for the return can also be specified. After all conditions are met then the user can submit the return which can be shown in figure 5.4(b).

Figure 4.4(b) Inventory Manager Vendor Returns

Products for Return Select All

Check boxes next to products you wish to return and specify quantities.

	PRODUCT	CONTAINER	SHIPPED	RETURN	CONDITION	REASON	NOTES
☒	Cheese Itz	(L1) Box	1	1	Damaged	Incorrect	Notes
☒	Cheese Itz	(L2) Unit	1	1	Used	Incorrect	Notes
☐	Eggo	(L1) Crate	1	0	Damaged	Other	Notes
☐	Eggo	(L2) Unit	1	0	Damaged	Other	Notes
☐	Pop Tarts	(L1) Box	1	0	Damaged	Other	Notes
☐	Pop Tarts	(L2) Unit	1	0	Damaged	Other	Notes

Figure 4.4(b)Inventory manager product selection for vendor returns

4.5 Inventory Manager Vendor RMAs

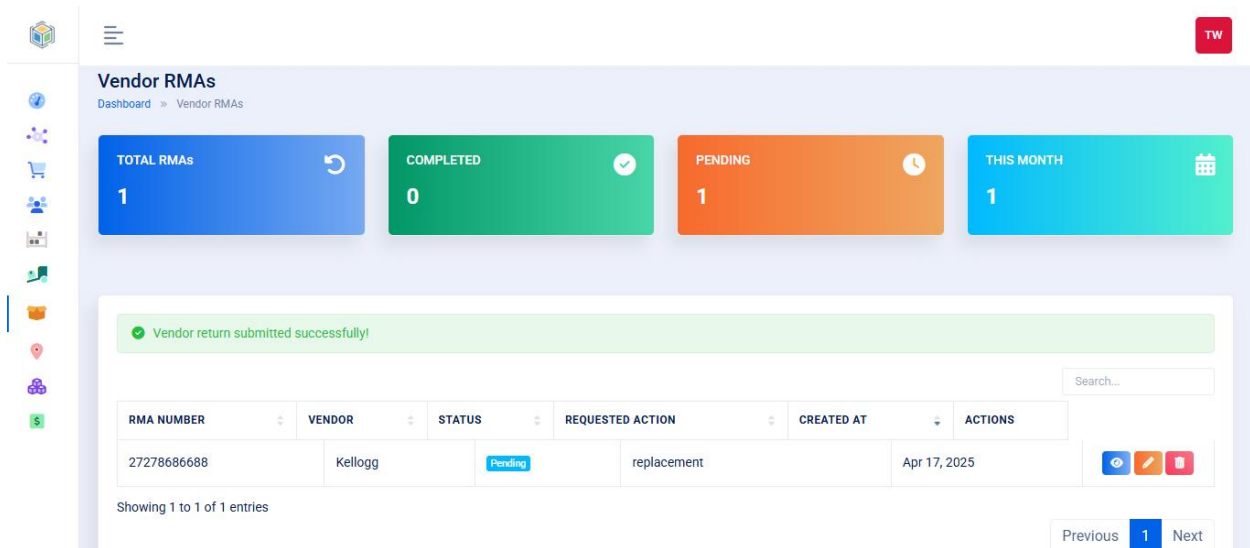


Figure 4.5 Inventory manager vendor RMAs page

This figure displays our system's Vendor RMAs which has information about completed, pending and total RMAs that are in the system. On this page the inventory manager can view, edit and delete vendor RMAs.

4.6 Inventory Manager Customer Returns

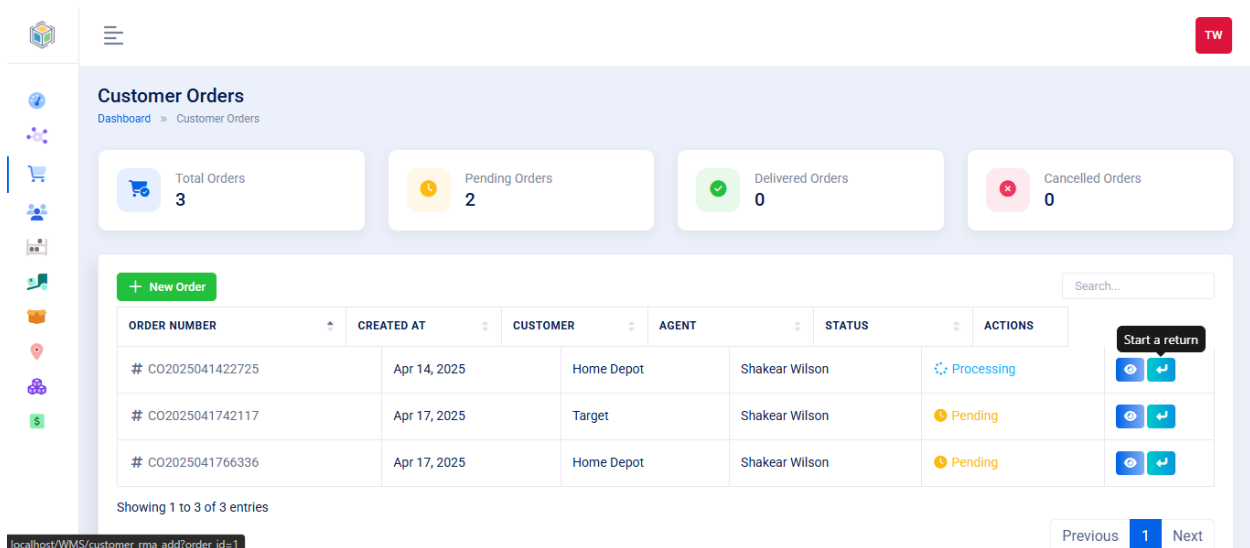


Figure 4.6 Inventory manager customer orders page

This page shows information about customer orders like total orders, pending orders delivered or cancelled orders. In this figure the inventory manager will need to navigate to the customer orders page to start a return for that customer's order. The inventory manager can also view and create customer orders.

4.7 Inventory Manager Customer Returns

The screenshot displays the 'New Customer Return' interface. At the top, there's a navigation bar with a 'TW' logo and a 'Back to RMAs' button. Below this, the page title 'New Customer Return' is followed by a breadcrumb trail: 'Dashboard > Customer RMAs > Create'. The main content area is divided into two sections. The top section, titled 'Customer: Home Depot', displays the 'Order #: C02025041422725' and a 'View Order' button. The bottom section, titled 'RETURN DETAILS', contains several input fields: 'RMA Number *' with the value 'RMA-250417-9508', 'Return Reason *' with a dropdown menu showing 'Damaged in Transit', 'Requested Action *' with a dropdown menu showing 'Refund', 'Shipping Carrier' with a dropdown menu showing 'FedEx', and 'Tracking Number' with the value '86626126354'. There is also a 'Notes' section at the bottom.

Figure 4.7 Inventory manager customer returns page

This page is just like the vendor returns page with the information needed being the requested action, return reason and the selected products that will be returned, the user can also include the tracking number and shipping carrier. Some bilipid information is the customer and the customer's order number. The user can also click on view order to view the customer's order that they are conducting a return on. The requested actions can be a replacement product, a refund or store credit. The products also have a drop down for their conditions which can be

used, damaged or defective, the reason for the return can also be specified. After all conditions are met then the user can create a customer RMA which can be shown in figure 5.7(b).

4.7(b) Inventory Manager Customer returns

Return Products

☐	PRODUCT	SKU	ORDER QTY	RETURN QTY	CONDITION	REASON
☒	Cheese Itz - Retail	9533711312	1	1	Like New	Not as Described
☐	Cheese Itz - Retail	9533711312	1	0	New/Unopened	Defective
☒	Eggo - Retail	4312646869	1	1	New/Unopened	Wrong Item
☐	Eggo - Retail	4312646869	1	0	New/Unopened	Defective

[Create Customer RMA](#)

WMS Project, 2025. Credits

Figure 4.7(b) Inventory Manager customer returns page

4.8 Inventory Manager Customer RMAs

Customer RMAs

Dashboard » Customer RMAs

Search...

RMA #	CUSTOMER	RETURN REASON	STATUS	CREATED	ACTIONS
RMA-250417-6876	Home Depot	Damaged	Requested	Apr 17, 2025 Tshakear Wilson	
RMA-250417-9508	Home Depot	Damaged	Requested	Apr 17, 2025 Tshakear Wilson	

Showing 1 to 2 of 2 entries

Previous **1** Next

WMS Project, 2025. Credits

Figure 4.8 Inventory manager customer RMAs page

This figure displays our system's Customer RMAs page which has information on RMA numbers, the customer, the status and the date in which it was created. On this page the inventory manager can view, edit and delete customer RMAs.

Inventory Manager Conclusion

The inventory manager also has access to storage & products pages that include vendor and customer catalog as well as vendor and customer containers. An inventory manager's work will not involve these pages, but they will still need access to them to view their data. They will only be able to view the page and will not be allowed to edit or add or delete these pages, the functionality and structure of these pages will be displayed in other roles.

Purchasing manager

A purchasing manager in a warehouse plays a critical role in ensuring that the facility is stocked with the right materials, products, and supplies needed to operate efficiently. While they aren't usually involved in the physical handling of inventory on the warehouse floor, their work directly influences everything from inventory levels to vendor relationships, lead times, and costs.

5.1 Purchasing manager

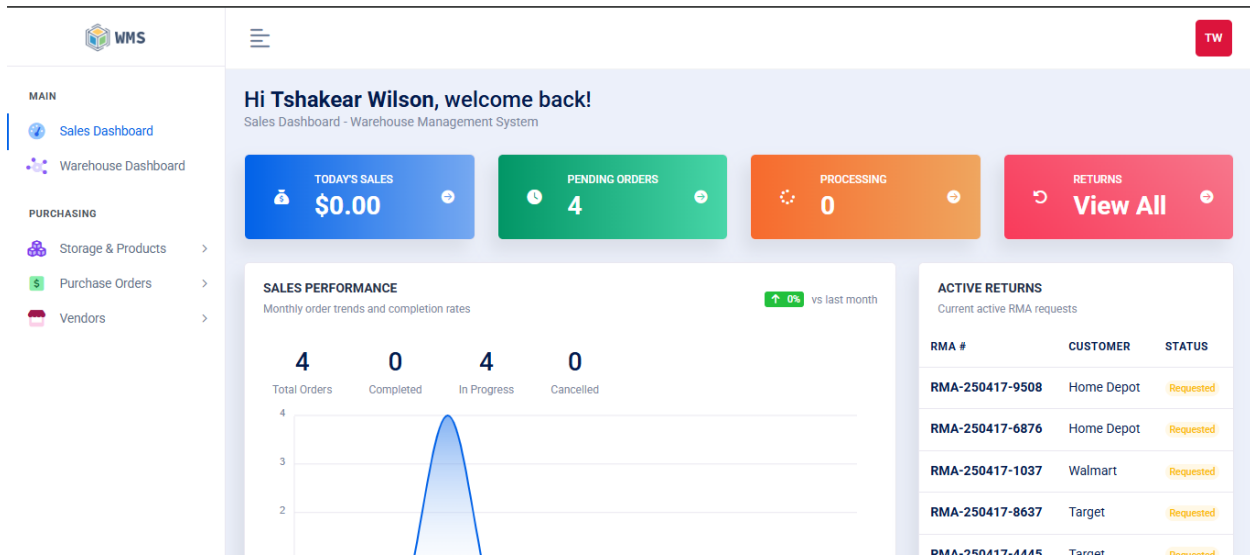


Figure 5.1 Purchasing managers dashboard

Figure 5.1 displays all the pages that a purchasing manager would access to. Purchasing manager's permissions consist of a dashboard, storage & products, purchasing orders and vendor. These are the only pages a purchasing manager would need to effectively help the warehouse.

5.2 Purchasing Manager's Vendors

Name
Ollie's

Address Line 1
William Penn

Address Line 2

City
Monroeville

State
Pennsylvania

Postal Code
15234

Country
United States

Figure 5.2 Purchasing managers add vendor's page

Purchasing managers are responsible for creating new connections with vendors so that the warehouse has someone to order products from. In figure 5.2 the manager would need to enter the following information from the vendor, the name of the vendor, address, city, state, postal code and the country the vendor is from. Once all the data is entered properly and submitted their vendor will then be added to the warehouse vendors, shown in figure 5.2(b).

5.2(b) Purchasing Manager Vendors

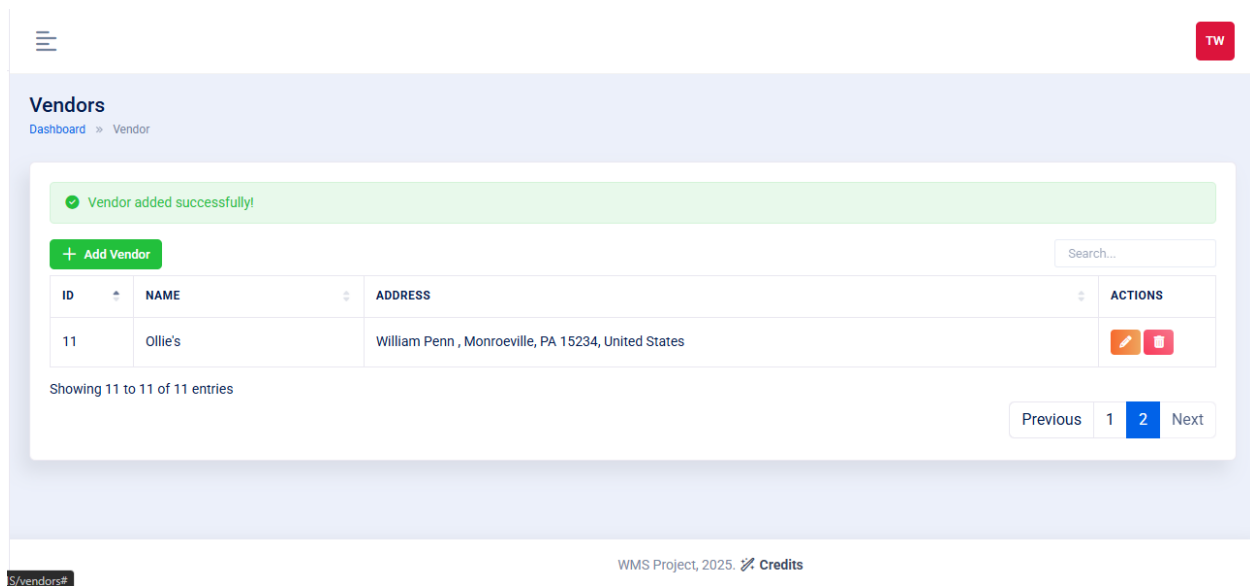
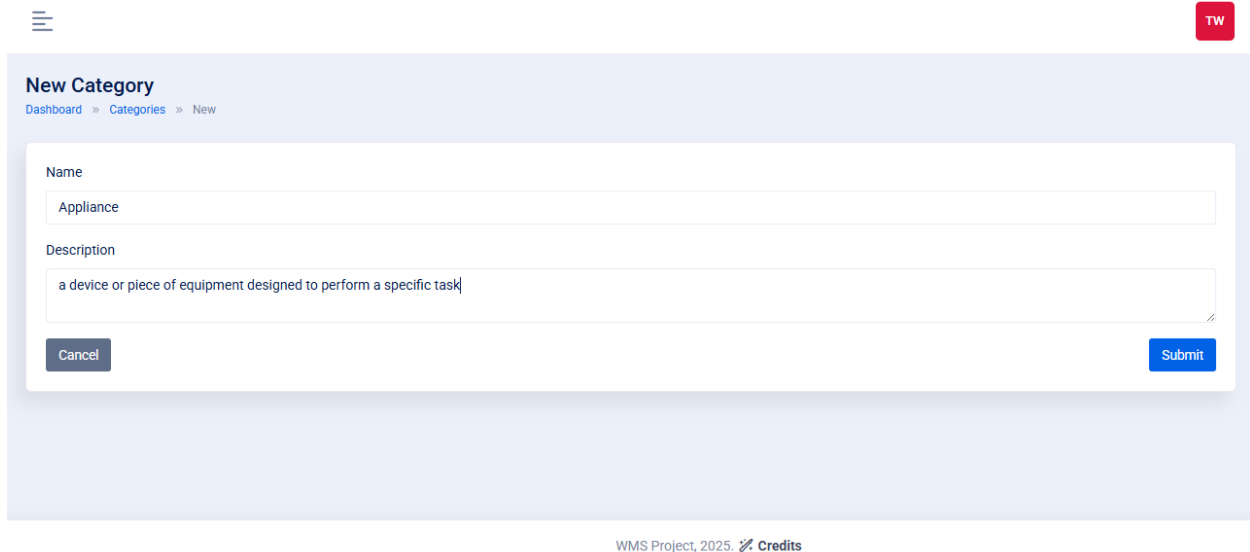


Figure 5.2(b) Purchasing manager vendor's page

5.3 Categories



New Category

Dashboard » Categories » New

Name

Appliance

Description

a device or piece of equipment designed to perform a specific task

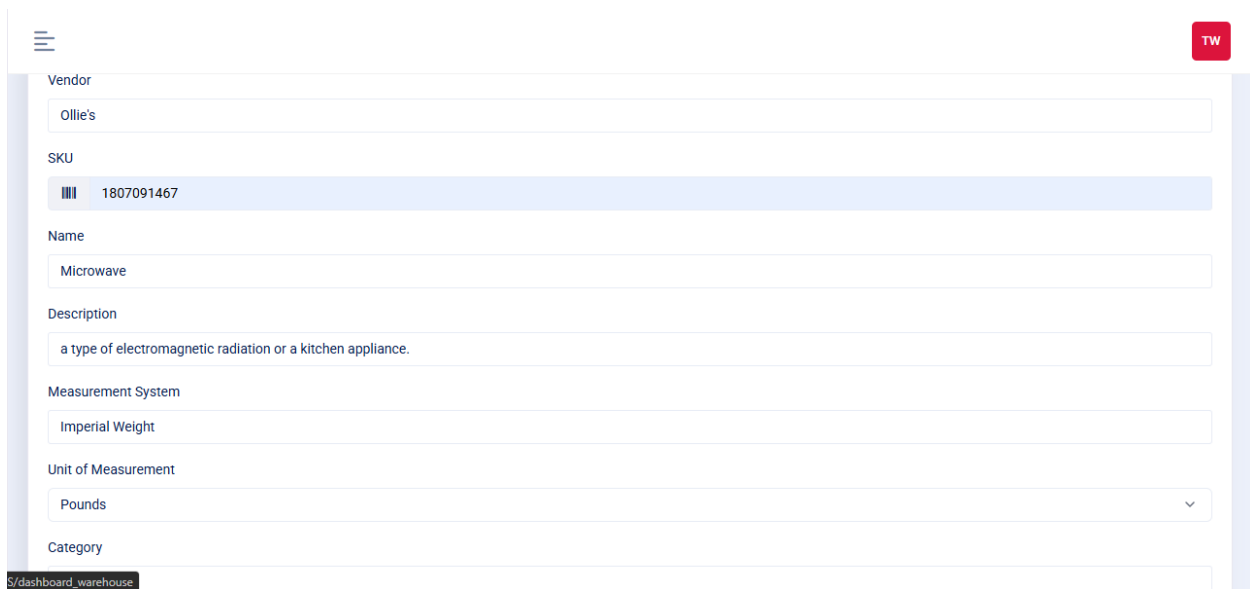
Cancel Submit

WMS Project, 2025. [Credits](#)

Figure 5.3 Purchasing manager categories add page

Figure 5.3 shows a basic page that requires a name and description to make a category that is going to be assigned to products that the manager buys from its vendors.

5.4 Purchasing Manager Vendor Catalog



Vendor

Ollie's

SKU

1807091467

Name

Microwave

Description

a type of electromagnetic radiation or a kitchen appliance.

Measurement System

Imperial Weight

Unit of Measurement

Pounds

Category

S/dashboard_warehouse

Figure 5.4 Purchasing manager vendor catalog add page

This page is crucial for a purchasing manager as this is the page that will add new products for the warehouse to order from. As seen in figure 5.4 we need a vendor, SKU, product name, measurements for set product and a category. In this example we'll use the vendor made in figure 5.2(b) and the category from figure 5.3. Once submitted, that product is now a part of vendor products shown in figure 5.4(b).

5.4(b) Vendor Products

The screenshot displays the 'Vendor Products' page in a web application. At the top, there is a navigation bar with a menu icon on the left and a red 'TW' logo on the right. Below the navigation bar, the page title 'Vendor Products' is shown, followed by a breadcrumb trail 'Dashboard » Vendor Products'. A green success message banner at the top of the main content area reads 'Vendor product added successfully!'. Below this, there are two buttons: '+ Add Product' (green) and 'Scan Barcode' (blue). To the right of these buttons is a search input field containing the text 'microwave'. The main content area features a table with the following columns: NAME, SKU, UNIT OF MEASUREMENT, VENDOR NAME, ACTIVE, and ACTIONS. The table contains one entry for a 'Microwave' with SKU '1807091467', unit of measurement 'Imperial Weight (Pounds)', vendor 'Ollie's', and status 'Active'. The 'ACTIONS' column for this entry contains two icons: a pencil (edit) and a trash can (delete). Below the table, a status message reads 'Showing 1 to 1 of 1 entries (filtered from 18 total entries)'. At the bottom right of the table area, there are pagination controls with 'Previous', '1' (selected), and 'Next' buttons. At the very bottom of the page, a small footer reads 'WMS Project, 2025. Credits'.

NAME	SKU	UNIT OF MEASUREMENT	VENDOR NAME	ACTIVE	ACTIONS
Microwave	1807091467	Imperial Weight (Pounds)	Ollie's	Active	 

Figure 5.4(b) Vendor products page

5.5 Purchasing Manager Vendor Containers

The screenshot displays a web application interface for adding vendor containers. It is divided into two main sections: a form on the left and a visual hierarchy on the right.

Form Section:

- Product Information:** A dropdown menu for 'Product' is set to 'Eggo'.
- Container Details:**
 - 'Parent Container' dropdown is set to 'Pallet'.
 - 'Container Type' dropdown is set to 'Box'.
 - 'Container Description' text area is empty.
- Quantity & Pricing:**
 - 'Quantity per container' is set to '10'.
 - 'Purchase price per container' is set to '\$ 82.50'.
 - 'Reorder Threshold' is set to '5'.
 - 'Active Container' toggle is turned on.

Visual Container Hierarchy:

- Level 0:** Product (Eggo)
- Level 1:** Pallet (No details, Qty: 1)
- Level 2:** New Box (Your new container, Qty: 10)

At the bottom of the form, there are 'Cancel' and 'Add Container' buttons.

Figure 5.5 Purchasing manager vendor containers add page

The purchasing manager now must put products into containers to make the correlation between the product, its price and the container that the warehouse will be receiving. Figure 5.5 shows how to make a product container by requiring the user to input the product, the container type quantity, price and reorder threshold. On the right-hand side is the systems visual container

hierarchy to help users visualize the containers. These products containers will be how the warehouse orders products on its purchase order page.

5.6 Purchasing Manager Orders

The screenshot displays the 'Purchasing Manager Orders' interface. It features a table of order items, an approval status section, and an actions section.

PRODUCT	UNIT PRICE	QUANTITY	TOTAL
1/2 Rebar (Level 1) Crate	\$1,000.00	10	\$10,000.00
1/2 Rebar (Level 2) Bundle	\$50.00	3	\$150.00
Steel Duct (Level 1) Tray	\$700.00	7	\$4,900.00
Steel Pipe (Level 1) Bundle	\$500.00	5	\$2,500.00
Steel Plate (Level 1) Pallet	\$2,000.00	2	\$4,000.00
Total:			\$21,550.00

Approval Status

Not Approved

Select Partner

Michael James (mJames@gmail.com)

☐ Send invoice PDF to partner

☐ CC order approver

Actions

Approve PO

Figure 5.6 Purchasing managers invoice approval page

The purchasing manager also deals with purchasing orders which you can refer to figure 4.3 for that page. On that page the user will choose a vendor, pick the products the warehouse needs and the quantity. After all data is entered the order can be submitted and the purchasing manager needs to approve the order. Figure 5.6 shows the purchase orders invoice where the manager can approve the order.

5.7 Purchasing Orders

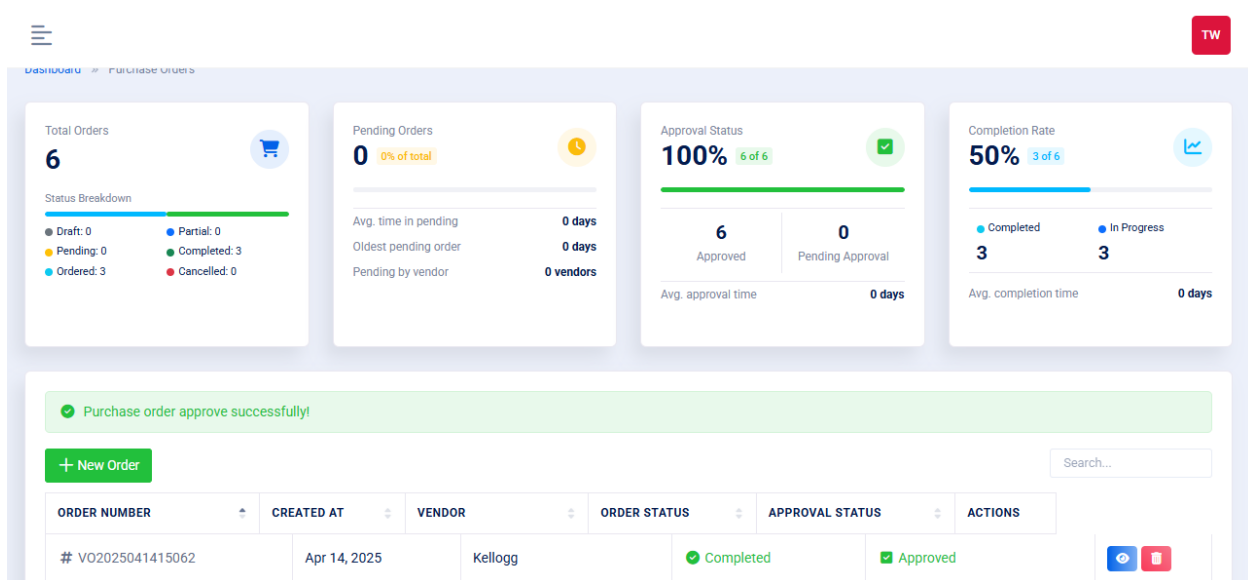


Figure 5.7 Purchasing orders base page

Figure 5.7 is the base page of purchasing orders. This page has special cards to demonstrate the status of orders, total orders, pending orders and completion rate. These are just small amounts of information the user can use to infer what to complete next. This page is how the purchasing manager gets to the purchasing orders invoice to approve orders as shown in figure 5.6.

Purchasing Manager Conclusion

The purchasing manager also has access to the customer catalog and container which look like the vendor pages, but since a purchasing manager does not directly work with customer products, they can only view the pages.

Shipping & Receiving Manager

A shipping and receiving manager oversee the critical entry and exit points of goods, essentially managing everything that comes into the warehouse and everything that goes out. Their role is hands-on, fast-paced, and central to keeping warehouse operations running smoothly and on schedule.

6.1 Shipping & Receiving Manager

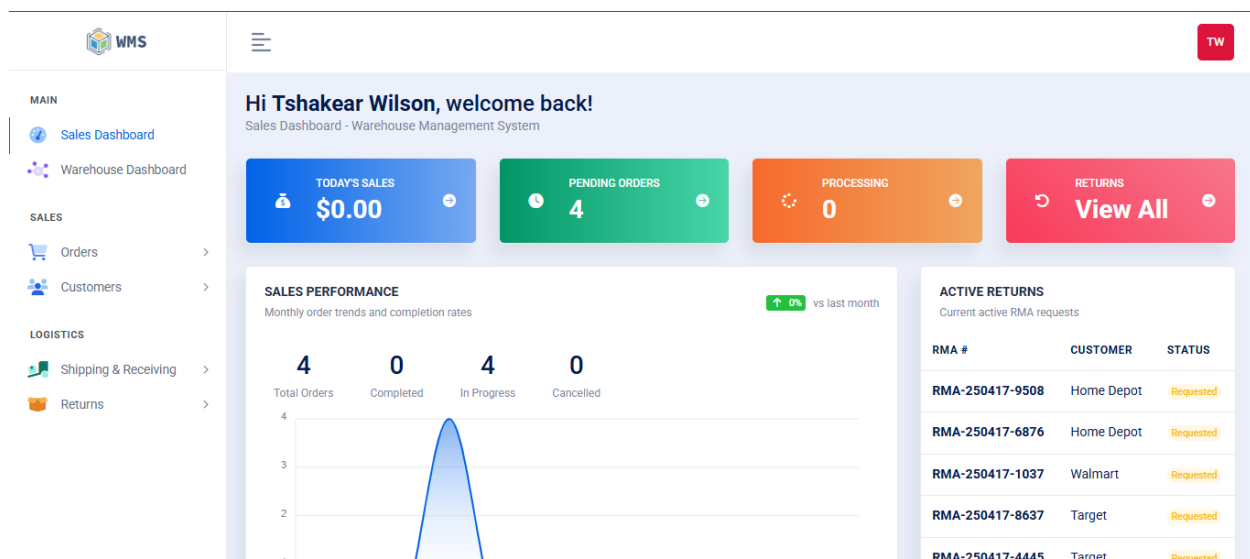


Figure 6.1 Shipping & receiving manager dashboard

A shipping and receiving manager will have access to orders which include customer orders, customers, shipping & receiving being incoming and outgoing shipments and returns. For returns the user can refer to figure 4.4 and 4.7, these figures show both returns. Incoming and outgoing shipments can be seen in figures 3.4(a) and 3.5(a), demonstrating the process of incoming products and the picking and packing process for customer orders.

6.2 Customer Orders

INVOICE

Order #C02025041422725

Mon, Apr 14, 2025, 10:30AM

Pending

Customer

Home Depot

Agent

Shakear Wilson

Invoice Date

Apr 14, 2025

Order Items

PRODUCT	UNIT PRICE	QUANTITY	TOTAL
Cheese Itz - Retail (Level 1) Box	\$70.00	1	\$70.00
Cheese Itz - Retail (Level 2) Unit	\$5.00	1	\$5.00

Order Summary

Subtotal

\$308.80

Tax (0%)

\$0.00

Total

\$308.80

Order Notes

N/A

This page holds general information about the customer's order and can be accessed by the shipping & receiving manager. The information on the page consists of the order number, the agent that made the order, the ordered items and an order summary.



TW

Name

Best Buy

Address

Mall Circle Drive

City

Monroeville

State

Pennsylvania

Zip Code

15146

Email

bestbuyguy@gmail.com

Agent

WMS

TW

Email
bestbuyguy@gmail.com

Agent
Shakear Wilson

Country Code
United States (+1)

Phone Number
(412) 555-5555

Tax Exempt
No

Cancel Add Customer

Figure 6.4 Customer's add page

Figure 6.4 shows the page where the shipping and receiving manager will create customers that the warehouse will have connections with selling products. The required information for a customer is their name, address, city, state, zip code, valid email address and phone number, an agent and if the business is tax exempt.

6.5 My customers

WMS

TW

MAIN

- Sales Dashboard
- Warehouse Dashboard

SALES

- Orders
- Customers
 - My Customers
 - Add Customer
 - Manage Customers

LOGISTICS

- Shipping & Receiving
- Returns

My Customers

Dashboard » Customers » My Customers

+ Add New Customer Search my customers

Best Buy
EZGAWONNPR
Not Exempt

bestbuyguy@gmail.com
+1 (412) 555-5555
Mall Circle Drive, Monroeville, PA 15146

Email Call Edit

+ New Order

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Figure 6.5 My customer's page

As shown in figure 6.4 the customer was added and now can be viewed on my customers page or the manage customers' page shown in figure 6.5. This page can be used for the manager to contact their customers about upcoming products and sales.

Customer service representative

A Customer Service Representative manages order-related concerns, answers questions, and handles product returns. Their main responsibilities are focused on resolving customer problems, guaranteeing accurate order handling, and assisting with customer return authorizations when needed.

7.1 Customer Service Representative

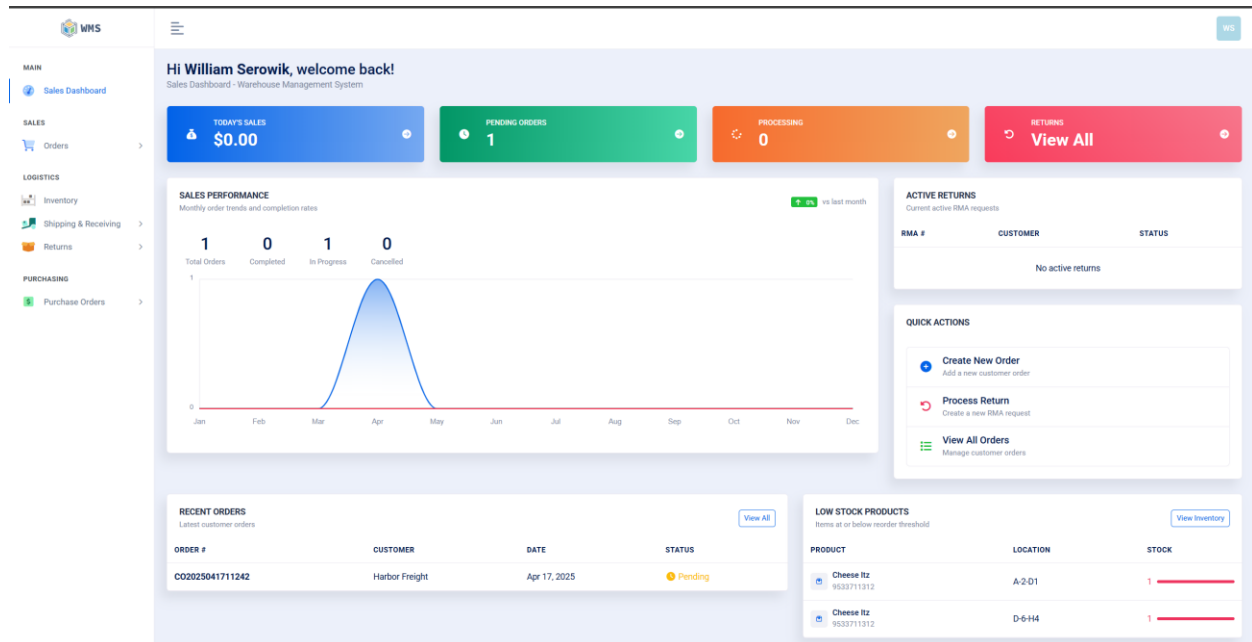


Figure 7.1 Customer representative's dashboard

Customer representatives have access to a limited number of pages necessary to handle customer operations. These include orders, customers, and returns. These permissions allow the Representative to review customer orders, manage return processes, and aid customers without access to the inventory or shipping workflows.

7.2 Customer service Representative – orders

As seen in figure 4.3 and 4.6 the representative can view order details such as status, products, quantities and any shipment details. They can use filters and a search function to quickly locate specific orders. While they cannot edit orders, they can provide updates or report issues to the fulfillment team.

7.3 Customer service Representative – Customers

Customers

Dashboard > Customers

+ Add Customer

NAME	EMAIL	PHONE	TAX EXEMPT	AGENT	ACTIONS
Harbor Freight # U0Z541WP1V	harborfreight@example.com	+1 (123) 123-1231	Not Exempt	WS William Serowik test-swiliam	
Home Depot # HUL840EXUL	homedepot@example.com	+1 (512) 555-9012	Exempt	TK Ty Kress tkress	
Lowes # UNY8XG2YGY	lowes@example.com	+1 (217) 555-1234	Exempt	SC Brian Colditz bubbycolditz	
Target # SKK455RFGK	target@example.com	+44 53245 345634	Not Exempt	WS William Serowik swiliam	
Walmart # MP2QDTIGKS	walmart@example.com	+1 (720) 555-5678	Not Exempt	TK Ty Kress tkress	

Showing 1 to 5 of 5 entries

Previous 1 Next

In figure 7.3 This page provides a list of registered customers in the system. It includes relevant details like company name and contact information. CSRs can access a customer's profile to assist with questions, view purchase history, and help start the return process if needed. They only have permissions for data review and not modifications

7.4 Customer Service Representative – RMAs (Return Requests)

In figure 4.4 you can see the processes for starting returns on customer orders. The information needed on this page is the product, the quantity, reason, reimbursement type, and additional customer notes. CSRs can either approve, reject, or leave as pending based on the warehouse's return policy. Status Changes are logged, and customers can be notified accordingly. The representative plays a key role in closing the feedback loop between the customer and warehouse

Customer Service Representative Conclusion

The representative does not interact directly with the physical inventory, locations, or purchase orders. Their role is strictly digital, and customer focused, access is limited mostly view only for most data. With some form-based inputs where customer action is required (such as return approvals). This design keeps customer information secure while giving the Representative the tools needed to support cases, returns, and inquiries accurately.

Sales Agent

A sales agent in a warehouse plays a critical role in the sale of products from the warehouse inventory to customers. While they aren't usually involved in the physical handling of inventory on the warehouse floor, their work directly influences everything from inventory levels to customer relationships, lead times, costs, and earnings.

8.1 Sales Agent

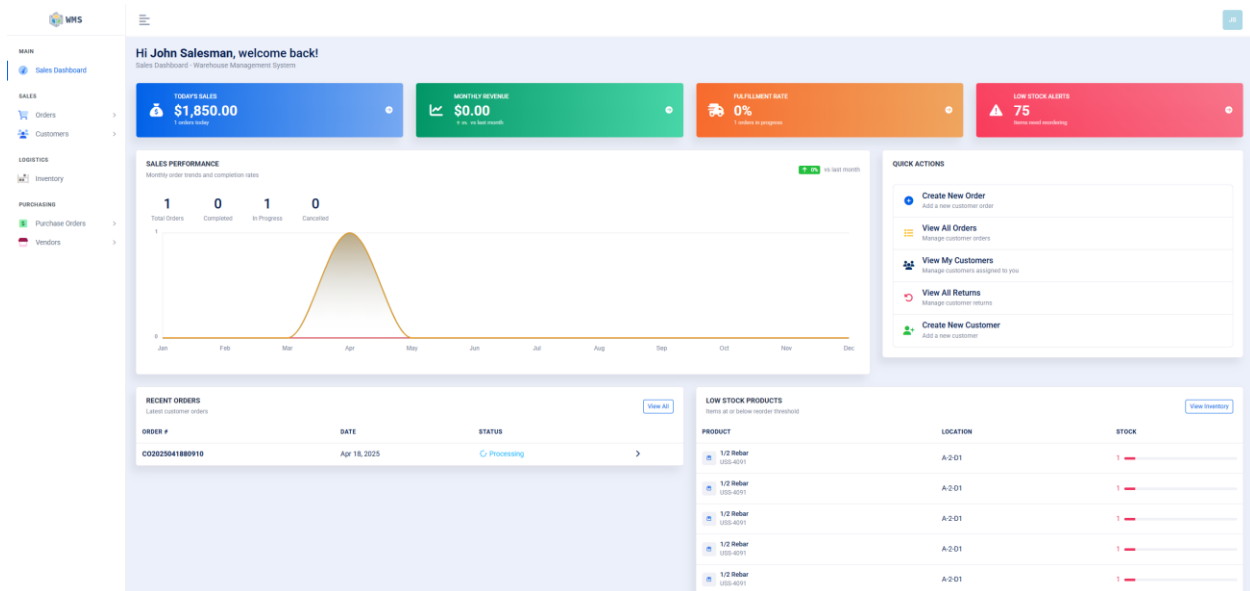


Figure 8.1 Sales Agent Dashboard

Figure 8.1 displays all the core pages that a sales agent would access. Sales agent permissions include the dashboard, orders, customers, limited access to purchasing orders, vendors, and inventory. These tools allow the sales manager to effectively oversee sales activity, manage customers, and ensure product availability. The dashboard provides real-time summaries of key metrics such as active orders, completed sales, customer activity, and inventory alerts. As shown in figure 8.1, the cards display high-level information that helps sales agents prioritize tasks and identify where follow-up may be needed.

8.2 Sales Agent's Customers

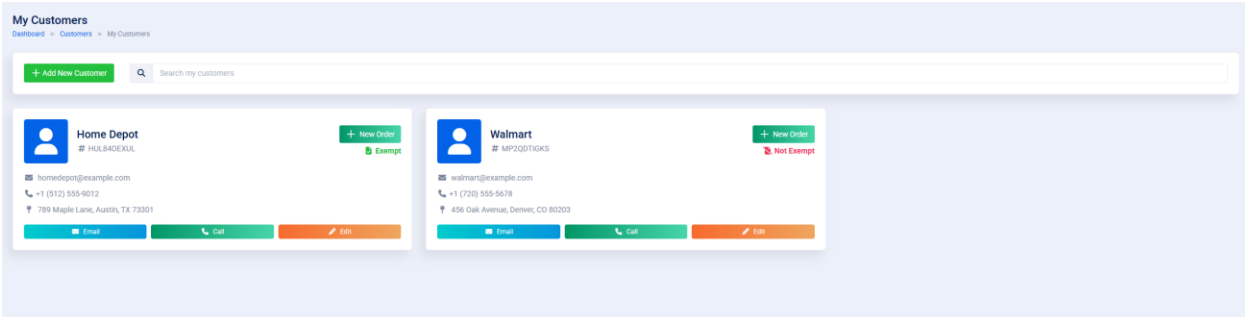


Figure 8.2 Sales Agent My Customers Page

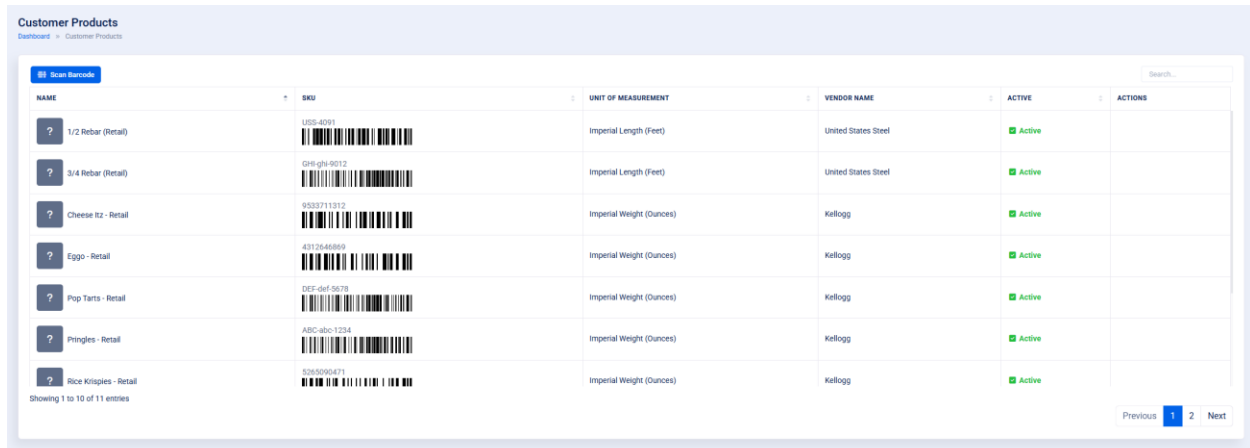
Figure 8.2 shows the view page for customers. This searchable tile list displays key details like contact info, address, and tax status. It helps the sales agent manage their customer relationships and respond to inquiries more effectively.

8.2(b) Sales Agent Add Customers

Figure 8.2(b) Sales Agent Add Customers Page

Sales agents are responsible for onboarding new customers. In figure 8.2(b), a new customer can be added by entering fields such as name, email, phone, billing and shipping addresses, and company name. Once submitted, the customer becomes selectable when creating new orders.

8.3 Sales Agent Customer Catalog



The screenshot displays the 'Customer Products' page in a sales agent interface. It features a table with the following columns: NAME, SKU, UNIT OF MEASUREMENT, VENDOR NAME, ACTIVE, and ACTIONS. The table lists six products, each with a question mark icon in the NAME column, a barcode in the SKU column, and a green 'Active' status in the ACTIVE column. The products are: 1/2 Rebar (Retail) from United States Steel, 3/4 Rebar (Retail) from United States Steel, Cheese Itz - Retail from Kellogg, Eggo - Retail from Kellogg, Pop Tarts - Retail from Kellogg, and Pringles - Retail from Kellogg. The table is paginated, showing 1 to 10 of 11 entries, with the current page being 2.

NAME	SKU	UNIT OF MEASUREMENT	VENDOR NAME	ACTIVE	ACTIONS
? 1/2 Rebar (Retail)	USS-4091 [Barcode]	Imperial Length (Feet)	United States Steel	Active	
? 3/4 Rebar (Retail)	USS-gh-0112 [Barcode]	Imperial Length (Feet)	United States Steel	Active	
? Cheese Itz - Retail	9533711312 [Barcode]	Imperial Weight (Ounces)	Kellogg	Active	
? Eggo - Retail	4313549889 [Barcode]	Imperial Weight (Ounces)	Kellogg	Active	
? Pop Tarts - Retail	DEF-d45-5678 [Barcode]	Imperial Weight (Ounces)	Kellogg	Active	
? Pringles - Retail	ABC-abc-1234 [Barcode]	Imperial Weight (Ounces)	Kellogg	Active	
? Rice Krispies - Retail	3333333333 [Barcode]	Imperial Weight (Ounces)	Kellogg	Active	

Showing 1 to 10 of 11 entries

Previous 1 2 Next

Figure 8.3 Sales Agent Customer Catalog Page

Figure 8.3 displays the page that allows the sales agent to browse and review products available to each customer. Products are organized by categories and customer-specific offerings. This helps ensure the sales agent only offers valid SKUs to each client. Inventory availability is also visible on this page

8.4 Sales Agent's Inventory View

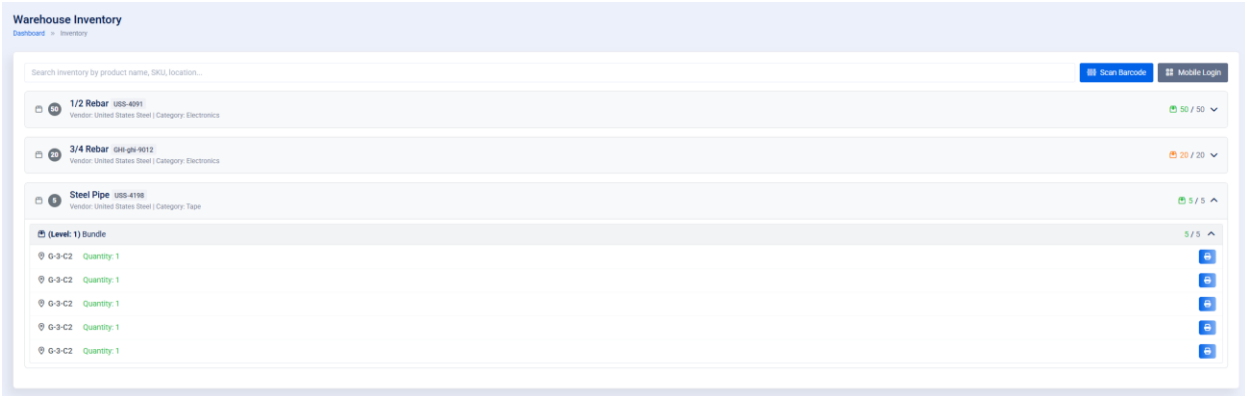


Figure 8.4 Sales Agent Inventory View Page

Though sales agents do not control warehouse stock levels, they need to verify inventory availability when creating or reviewing orders. Figure 8.4 shows a read-only view of inventory levels across all products, including quantities on hand and reorder thresholds. This allows for more accurate order placement and customer service.

8.5 Sales Agent View Orders

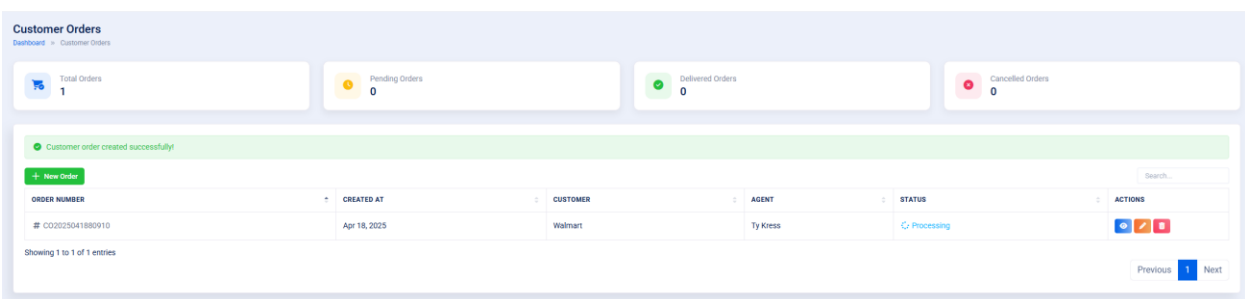


Figure 8.5 Sales Agent View Orders Page

Figure 8.5 displays the sales order view page, which provides a searchable list of all orders created by the agent or team. Sales agents can review statuses, shipping progress, and

order totals. This view helps with follow-ups and customer communication regarding current and past orders.

8.5(b) Sales Agent Add Orders

New Customer Order

Dashboard > Customer Orders > New Order

CUSTOMER INFORMATION

Customer: Walmart Tax Rate: 6

ORDER PRODUCTS

PRODUCT	CONTAINER	COST	QTY	TOTAL
1/2 Rebar (Retail)	(Level 2) Bundle (Available: 50)	70.00	10	\$700.00
Steel Pipe (Retail)	(Level 1) Bundle (Available: 5)	\$75.00	2	\$1,150.00

ORDER SUMMARY

Subtotal: \$1,858.00
Tax (6%): \$111.00
Total: \$1,969.00

ORDER DETAILS

Agent: Ty Kress
Order Date: 2025-04-18
Order Status: Processing
Order Notes: Enter any additional notes about this order...

Figure 8.5(b) Sales Agent Add Orders Page

Sales agents are responsible for creating and managing sales orders. In figure 8.5(b), the agent selects a customer and chooses from available inventory. The system allows them to enter product quantities, pricing (if applicable), and additional order notes. Once submitted, the order appears in the order history, and its status can be updated as it progresses through fulfillment.

8.6 Sales Agent Vendors

Vendors
Dashboard > Vendor

[+ Add Vendor](#)

ID	NAME	ADDRESS	ACTIONS
1	3M	3M Center 90th Street, Maplewood, MN 55109, United States	
2	United States Steel	600 Grant Street U.S. Steel Tower, Pittsburgh, PA 15219, United States	
3	PPG Industries	PPG Place 300 Fifth Avenue, Pittsburgh, PA 15222, United States	
4	Kraft Heinz	200 East Randolph Street Suite 300, Chicago, IL 60601, United States	
5	Scotch	3M Center 90th Street, Maplewood, MN 55109, United States	
6	Apple Inc	One Apple Park Way, Cupertino, CA 95014, United States	
7	Kellogg	1 Kellogg Square, Battle Creek, MI 49017, United States	
8	Herman Miller	1000 Furniture Drive, Zeeland, MI 49484, United States	
9	Frito-Lay Inc	One Frito-Lay Drive, Plano, TX 75024, United States	
10	Dell Technologies	One Dell Way, Round Rock, TX 78682, United States	

Showing 1 to 10 of 10 entries

Previous [1](#) Next

Figure 8.6 Sales Agent View Vendors Page

Sales agents have read-only access to vendor information. Figure 8.6 displays a basic vendor view showing names, contact details, and associated products. This information can assist in communicating product details to customers or coordinating with purchasing when stock needs to be replenished.

8.7 Sales Agent View Purchase Orders

Purchase Orders
Dashboard > Purchase Orders

Total Orders
2

Status Breakdown

- Draft: 0
- Pending: 0
- Ordered: 1
- Partial: 0
- Completed: 1
- Cancelled: 0

Pending Orders
0 of total

Avg. time in pending
Oldest pending order
Pending by vendor

0 days
0 days
0 vendors

Approval Status
100% 1 of 2

2 Approved
0 Pending Approval

Avg. approval time
0 days

Completion Rate
50% 1 of 2

1 Completed
1 In Progress

Avg. completion time
0 days

[+ New Order](#)

ORDER NUMBER	CREATED AT	VENDOR	ORDER STATUS	APPROVAL STATUS	ACTIONS
# V02025041635277	Apr 16, 2025	United States Steel	Completed	Approved	View
# V02025041682469	Apr 16, 2025	United States Steel	Ordered	Approved	View

Showing 1 to 2 of 2 entries

Previous [1](#) Next

Figure 8.7 Sales Agent View Purchase Orders Page

Purchase Order #VO2025041635277

Dashboard

Purchase Orders

Order Overview

Order Date

Apr 16, 2025

Vendor

United States Steel

Order Items

Product	Unit Price	Quantity	Total
Steel Duct (Level 1) Tray	\$700.00	1	\$700.00
Steel Duct (Level 2) Unit	\$25.00	1	\$25.00
Total:			\$725.00

Order Notes

N/A

Status

Completed

Approval Status

Approved

William Serowik

Partner Information

Michael James

m.james@gmail.com

Actions

Figure 8.7(b) Purchase Order Overview

Sales agents can also view purchase orders to track incoming stock and estimated delivery dates. As seen in figure 8.7(b), they can check vendor names, items ordered, quantities, and statuses. This page is for viewing only, helping sales agents manage expectations and update customers accordingly.

Sales Agent Conclusion

The sales agent is essential to the customer's experience within the warehouse system. With access to customer and order creation tools, and visibility into vendors, inventory, and purchase orders, they can provide fast, accurate service. Their role bridges customer interaction and warehouse readiness, ensuring a smooth sales pipeline.

Warehouse Manager

The warehouse manager's role includes access to all the permissions and pages of the roles previously mentioned, see figure 9. In addition to the pages laid out previously, the warehouse manager has access to view User's information.

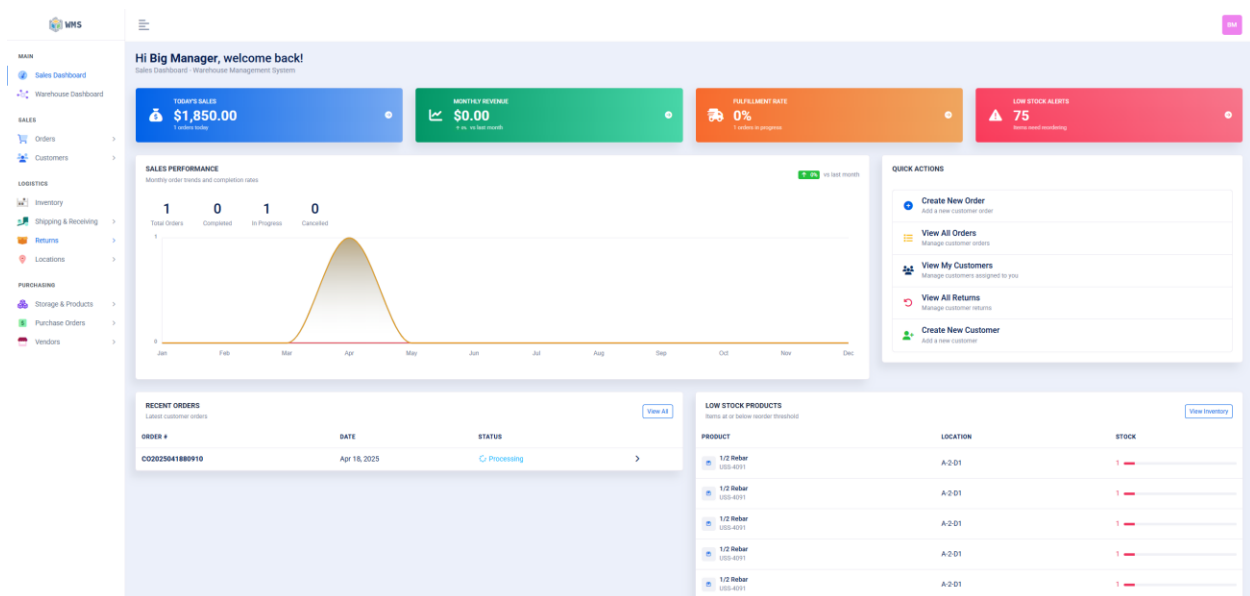


Figure 9 Warehouse Manager Dashboard

9.1 Warehouse Manager Users

Users

Dashboard > Users

+ Add User

ACCOUNT	EMAIL	ROLE	ACCOUNT STATUS	ACTIONS
Admin User admin	admin@wms.com	Administrator	Suspended	
Austin Oertel aoertel	austinoertel@gmail.com	Administrator	Activated	
Brian Colditz bubbycolditz	bubbycolditz@gmail.com	Administrator	Activated	
Clayton Sanner csanner	csanner024@penwest.edu	Administrator	Activated	
Shakeer Wilson swilson	swager1793@gmail.com	Administrator	Activated	
Ty Kress tkress	tykress@gmail.com	Administrator	Activated	
William Serowik wserowik	serowik.william@gmail.com	Administrator	Activated	

Showing 1 to 7 of 7 entries

Previous 1 Next

Figure 9.1 User View Page

Warehouse managers have read-only access to user accounts within the system.

Figure 9.1 shows a list of users and their associated roles. This allows the manager to verify

who is assigned to various responsibilities, helping them coordinate across departments more efficiently. However, user roles and permissions cannot be edited by this role.

Warehouse Manager Conclusion

The warehouse manager has the broadest operational access in the system, acting as the bridge between roles and ensuring all departments—from sales and customer service to receiving and inventory—operate in sync. With control over locations, audit cycles, and cross-department reports, the warehouse manager is positioned to oversee efficiency, accuracy, and daily execution across the facility.

References

Table of Index / Glossary

AJAX - stands for Asynchronous JavaScript and XML. It is a set of web development techniques using various technologies to create dynamic web applications

Apache – A widely used open-source web server software that helps deliver web content to users.

Docker – A platform that uses containers to run applications in isolated environments, ensuring consistent functionality across systems.

Location – A specific storage space in the warehouse defined by rack, bay, level, and position.

Mobile Login – A feature that allows warehouse associates to log in to the system using mobile devices for flexibility and efficiency.

PHP – The main programming language used in this system to manage server-side scripting and database interaction.

QR (Quick response) - a two-dimensional barcode that can be scanned with a smartphone to access information.

RMA (Return Merchandise Authorization) – A process that tracks and approves product returns from customers or to vendors.

RBAC (Role-Based Access Control) – A system of assigning permissions to users based on their job responsibilities.

SKU (Stock Keeping Unit) – A unique code or number used to identify individual products.

Appendix: Team Details and Individual Contributions

The work shown was distributed among group members who developed the Warehouse Management System with this document being led by Shakear Wilson. This document was started in the final phases of the project and was completed alongside the implementation of the software for our system. Each member of the team completed their section as well as helped to add to other members' sections. The document was created by all the following individual efforts:

Brian was responsible for Project overview and application, system block diagram, Project implementation details, team details along with the user's manual for this document. During the implementation phase of this project Brian was the leader in the group and helped develop web pages, tables for the database, the implementation of functionality and logic in our system, security/error checking web pages, the development of RBAC and optimizing code.

Ty was responsible for Project overview and application, system block diagram, user's manual, team details, challenges during implementation and the Glossary for this document. During the implementation phase of this project Ty helped in developing the functionality of many web pages, the implementation of functionality and logic in our system along with handling weekly reports.

William was responsible for Project implementation details, system block diagram, user's manual, challenges during implementation and motivation for this document. During the implementation phase of this project William helped in developing the functionality of many web pages, security/error checking web pages, the implementation of functionality and logic in our system and development of RBAC.

Shakear was responsible for project overview and application system block diagram, use of software engineering principles, difference from design document, comparison to existing similar products user's manual, challenges during implementation, team details and motivation for this document. During the implementation phase of this project Shakear helped in developing the functionality of many web pages, the implementation of functionality and logic in our system.

Other contributions that were coordinated by group members consisted of:

- Discord meetings/discussion
- In person discussion
- Formatting and planning outline of document
- Meeting at Writing Center
- Feedback and Proofreading

Appendix: Writing Center report

Appendix: Code listing

Appendix: Workflow Authentication

I, Brian Colditz, agree with the details defined in this document that represent functional requirements of WMS. Also, I agree that the work that was done as stated by this document

Signature: 

Date: 4/18/2025

I, Ty Kress, agree with the details defined in this document that represent functional requirements of WMS. Also, I agree that the work that was done as stated by this document

Signature: 

Date: 4/18/2025

I, Shakear Wilson, agree with the details defined in this document that represent functional requirements of WMS. Also, I agree that the work that was done as stated by this document

Signature: 

Date: Date: 4/18/2025

I, William Serowik, agree with the details defined in this document that represent functional requirements of WMS. Also, I agree that the work that was done as stated by this document

Signature: 

Date: 4/18/2025